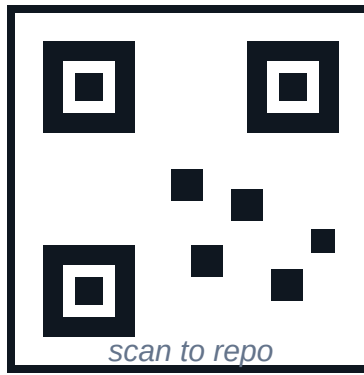


Measuring Neural-Network Robustness Under Adversarial Perturbation



Experimental Evaluation of Model Behavior Under Gradient-Based Attacks

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Security Engineering • AI / Agent Security • Cloud Security

MIT • Python 3.12 • HEAD 0fadaa2 • verified 2026-09-26

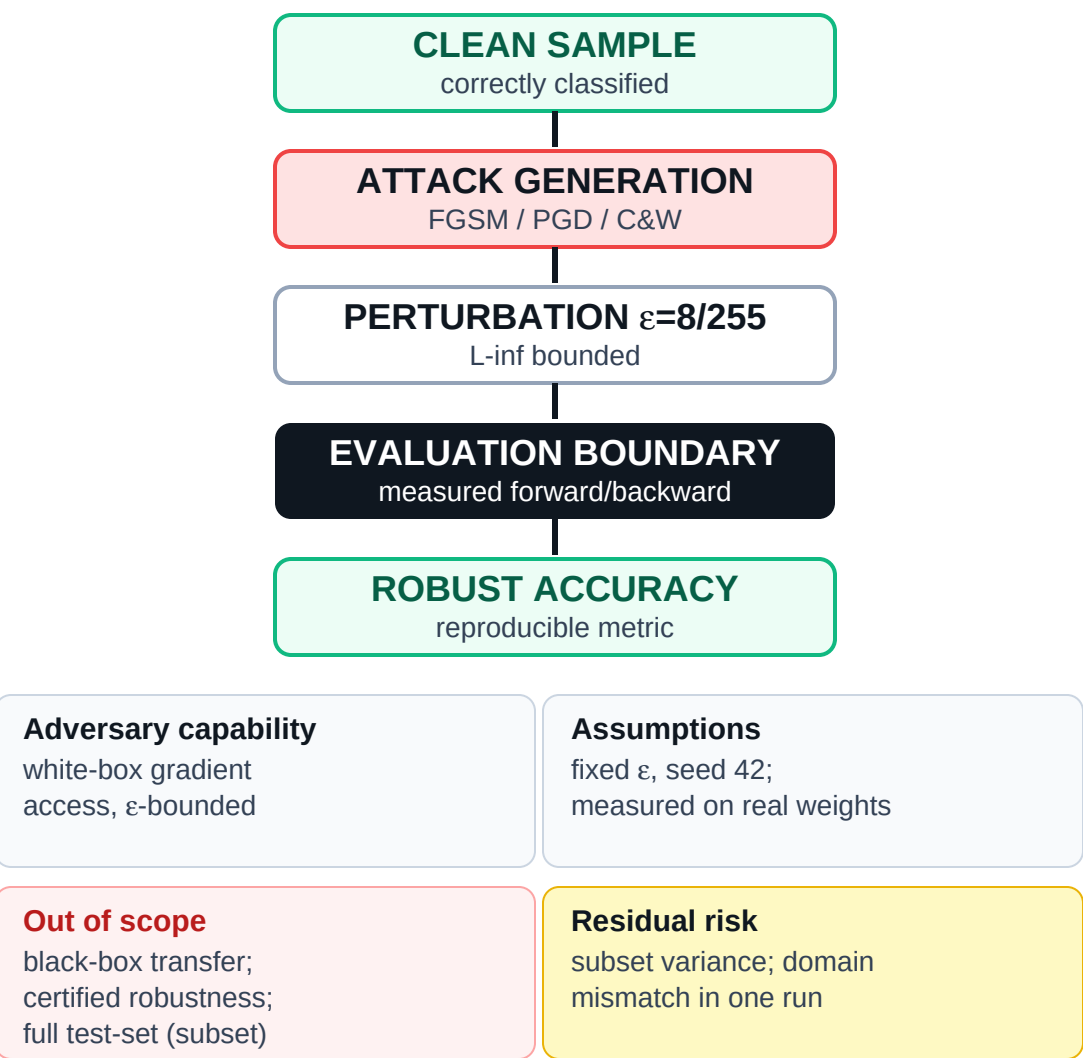
RESEARCH QUESTION

How does a model's accuracy collapse under FGSM/PGD/C&W attacks — and are the reported numbers actually measured, not projected from literature?

01 • THE FAILURE MODE

Small L-infinity perturbations flip confident model predictions. Reporting robustness is easy to fake: a single-step FGSM can look 'robust' due to gradient masking, and literature values can be pasted in as if measured. This lab separates REAL measured runs from clearly-labeled literature projections.

02 • THREAT MODEL

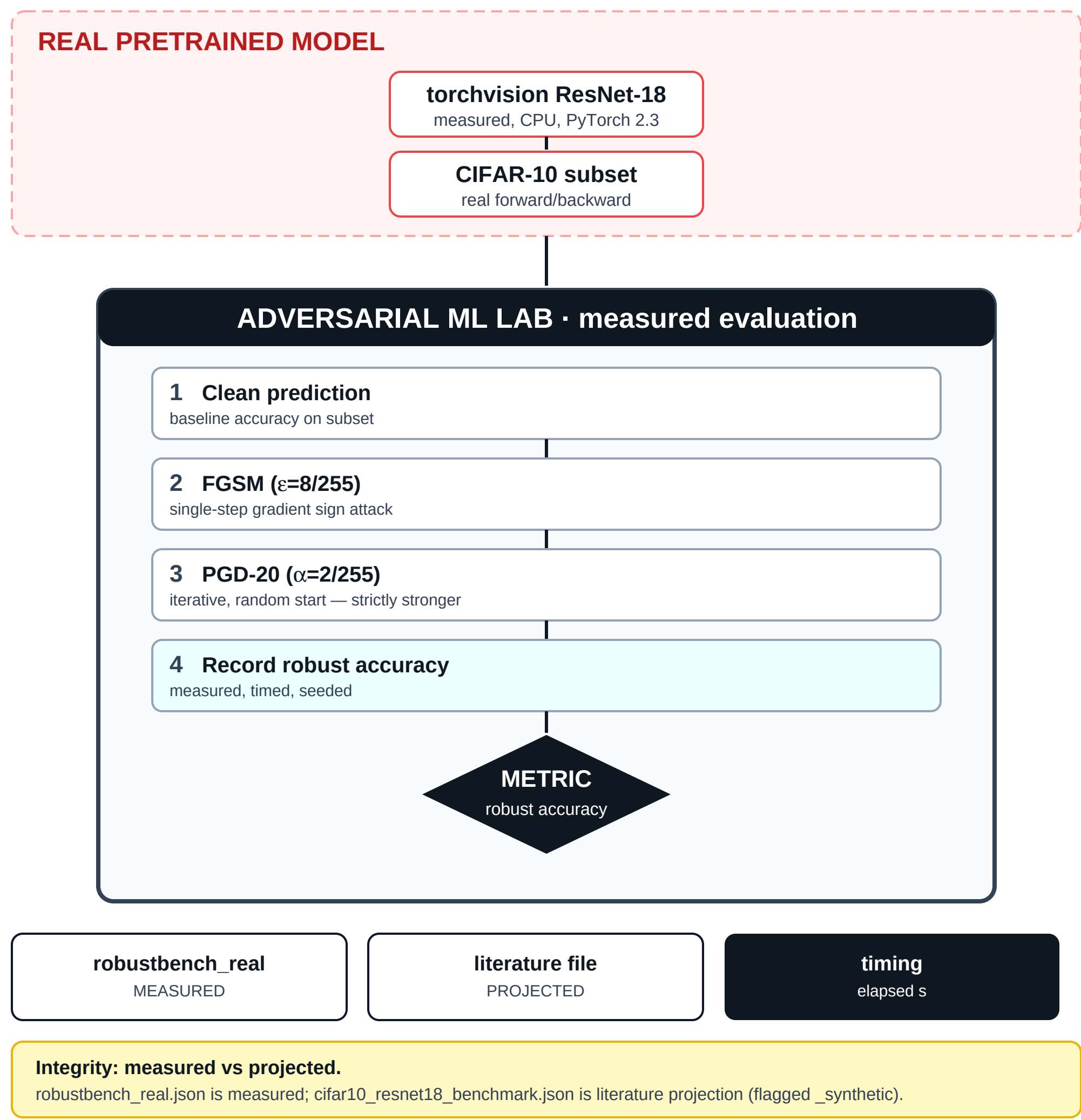


03 • OBJECTIVE

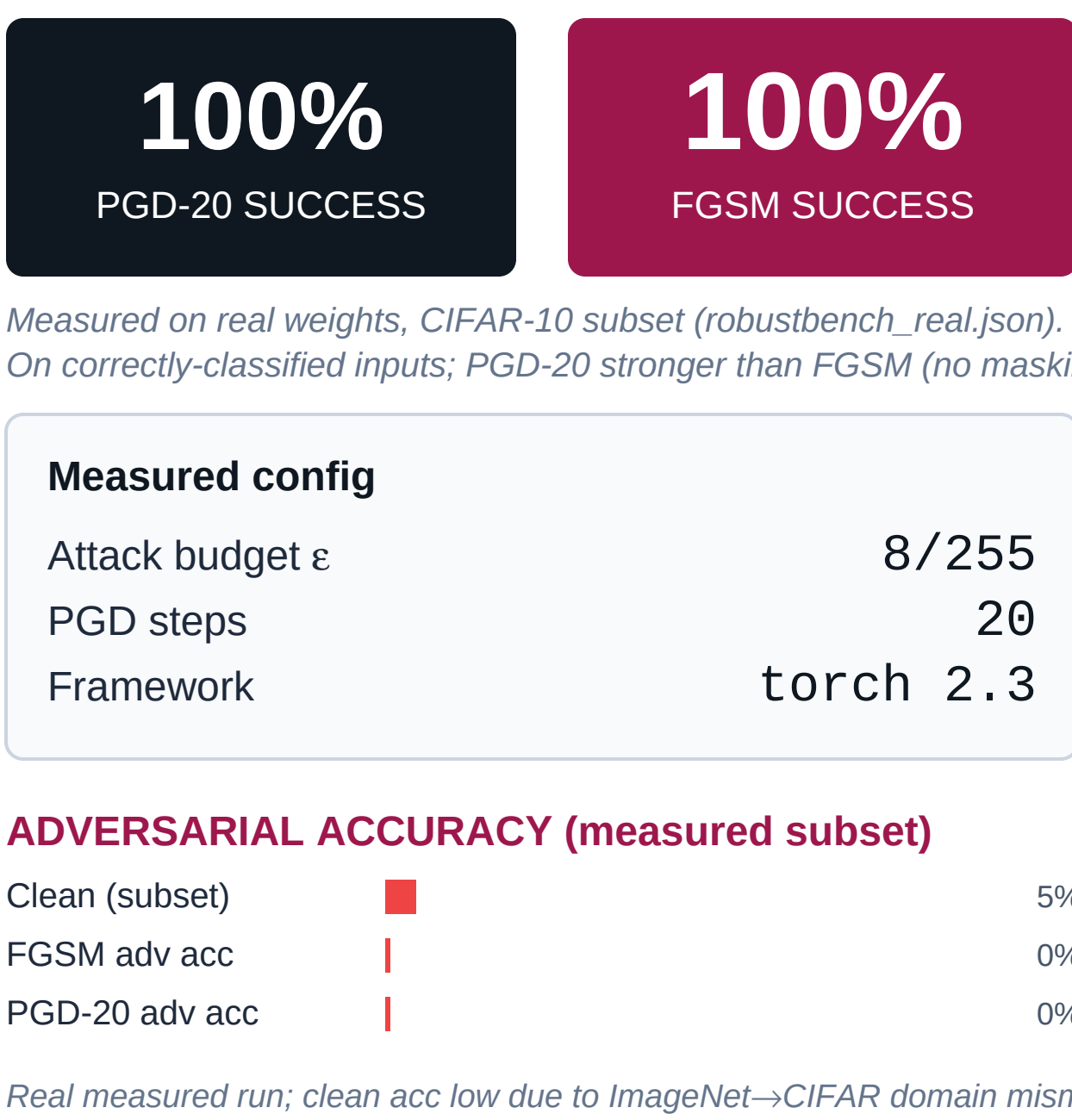
Measure real robustness collapse under gradient attacks and keep measured results strictly separate from literature projections.

- 01 – FGSM / PGD / C&W attack impls
- 02 – Measured runs on real weights
- 03 – Robust-accuracy metrics + timing
- 04 – Label projected numbers as projected

HERO • ADVERSARIAL EVALUATION LOOP



RESULTS • MEASURED ATTACK SUCCESS



LITERATURE PROJECTION (labeled)
cifar10_resnet18_benchmark.json is a LITERATURE PROJECTION (Madry 2018): undefended PGD ~0%, Madry-AT ~45%. NOT measured here — shown as context.

NOT ESTABLISHED BY THIS REPO
State-of-the-art robustness. Certified guarantees. Full-test-set numbers. Black-box transferability.

SECURITY BOUNDARY

PROTECTS / DETECTS - measures FGSM/PGD/C&W - real weights, real data - gradient-masking check - seeded + timed - measured vs projected	DOES NOT ESTABLISH - does not defend the model - no certified bound - no SOTA claim - subset, not full set
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Figure 1. Measured adversarial loop on real pretrained weights. Projected literature values are stored separately.

METHOD • ATTACK EVALUATION



MEASURED ≠ PROJECTED

CONTRIBUTION

A gradient-attack evaluation harness (FGSM/PGD/C&W) that measures robustness collapse on real pretrained weights and enforces a hard separation between measured results and literature projections — the projected file is explicitly flagged _synthetic and not reproducible.

LIMITATIONS

- 1. Measured run uses a small subset, not full test set.
- 2. Clean accuracy low (ImageNet→CIFAR domain mismatch).
- 3. Literature file is projection, not reproducible.
- 4. White-box only; no black-box/transfer attacks.
- 5. No certified-robustness claim.

FUTURE WORK

- Full 10k CIFAR-10 test-set measured run.
- CIFAR-trained model (remove domain mismatch).
- Madry adversarial-training measured baseline.
- AutoAttack ensemble evaluation.
- Certified-robustness comparison.

CONCLUSION

The lab measures real robustness collapse under gradient attacks and keeps measured results strictly apart from literature projections. On the tested subset, FGSM and PGD-20 reach 100% success. It evaluates robustness; it does not certify it.

REPRODUCE

```
python benchmark/robustbench_baseline.py
pytest tests/
```